

09/07/26

Assessment tool

Stack queue and IL GATE questions.

1) Stack:

push 54, push 52, pop (removes 52), push 55,
push 62, pop $\rightarrow S = 62$.

Queue:

enq 21, enq 24, deq (removes 21), enq 28, enq 32,
deq (removes 24) $\rightarrow q = 24$ $Star = 62 + 24 = 86$

2) Postfix: 10 5 + 60 6 / * 8 -

10 - 10

5 - 10 5

+ + 10 5 +

60 10 5 + 60

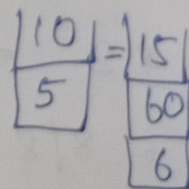
6 10 5 + 60 6

/ / 10 5 + 60 6 /

* * 10 5 + 60 6 / *

8 10 5 + 60 6 / * 8

- - 10 5 + 60 6 / * 8 -



$$10 + 5 = 15$$

$$60 / 6 = 10$$

$$15 * 10 = 150$$

$$150 - 8 = 142$$

enqueue(21) $\rightarrow$ [21]

enqueue(24) $\rightarrow$ [21, 24]

enqueue(28) $\rightarrow$ [24, 28, 32]

dequeue() $\rightarrow$ removes 21

enqueue(32) $\rightarrow$ [24, 28, 32]

$Q = \text{dequeue}() \rightarrow Q = 24$

$$S + Q = 62 + 24 = 86.$$

2) Postfix Expression

Expression : 10 5 +

3) It can implement both a stack and a queue efficiently.

Ex) Postfix Expression :

Expression : 8 2 3 ^ / 2 3 * + 5 1 * -

$$* 2^3 = 8$$

$$* 8/8 = 1$$

$$* 2 \times 3 = 6$$

$$* 1 + 6 = 7$$

After evaluating the first *,
Stack contains : 1, 6.

$$\Rightarrow 6, 1.$$

6) 504.
Expression : $a + b \times c$

* $\wedge$ is right associative.

* postfix : $abc * + def ^^ -$

$\Rightarrow abc + def ^^ - *$

7) stack PUSH/POP sequence :

PUSH(10)

PUSH(20)

POP $\rightarrow$ 20

PUSH(10)

PUSH(20)

POP $\rightarrow$ 20

POP $\rightarrow$ 10

POP $\rightarrow$ 10

PUSH(20)

POP $\rightarrow$ 20

$\Rightarrow 20, 20, 10, 10, 20$.

4) A language accepted by a pushdown automaton with REVERSE operation is more powerful than a context-free language.